SOFTWARE PRODUCT · DOCUMENTATION

Lalaba

Marketplace

A multi-role laundry marketplace connecting customers, laundromats, home washers, staff, fulfillment operations, and platform administration.

ROLE	PERIOD	PLATFORMS
Junior Full-Stack Engineer	**Oct 2025 – Aug 2026**	**Web · Mobile · Backend**

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1. Overview

A multi-role laundry marketplace connecting customers, laundromats, home washers, staff, fulfillment operations, and platform administration.

PROJECT	Lalaba
TYPE	Marketplace
ROLE	Junior Full-Stack Engineer
PERIOD	Oct 2025 – Aug 2026
PLATFORMS	Web · Mobile · Backend
TEAM	Built as part of a small engineering team

2. Scope and personal contribution

> Counts such as "114 Edge Functions" describe the whole platform, not only Prince's own work. His own contributions are the areas of ownership listed in this section.

- Branch-scoped device approval
- Wallet webhook idempotency
- Server-authoritative pricing
- Permission/role enforcement

What was built, in detail:

- Built out a NestJS + Apollo GraphQL backend (187 module directories, 57 resolvers, 68 Mongoose schemas) shared by both mobile apps and the admin panel, as part of a small engineering team.
- Implemented branch-scoped staff device approval — single-active-session gating, a 5-minute cached auth TTL, and per-branch grants enforced server-side, not just hidden in the UI.
- Built server-authoritative platform-fee, booking-availability, and booking-policy modules so pricing and capacity rules can't be recomputed client-side.
- Implemented the Xendit wallet top-up webhook: timing-safe callback-token verification, amount/currency validation against the stored intent, and a uniquely-indexed ledger write so a retried callback can never double-credit.

3. Product surfaces

Customer App

`Expo · React Native`

The app end customers use to find laundry providers, book pickup/delivery, and pay from an in-app wallet.

- Firebase Auth (JS SDK) + App Check
- Google Maps for provider discovery
- Custom fetch-based GraphQL client — no Apollo/codegen

Partner App

`Expo · React Native`

The app laundromats, home washers, and staff use — bookings, verification, and device-gated access.

- Camera + face detection + biometrics for verification/KYC
- Branch-scoped staff device approval
- QR-code support for pickup/handoff flows

Admin Panel

`Next.js`

Internal operations console — 29 feature routes mirroring the backend's own module boundaries.

- Merchants, orders, wallets, promotions, campaigns, booking-policy
- Hand-written GraphQL query/mutation layer (no client library)
- Role/permission management UI

Marketing Website

`Next.js`

A standalone public-facing site — intentionally has no live connection to the backend.

- Static marketing content only
- No auth, API, or payment logic
- Separate repo, separate deploy

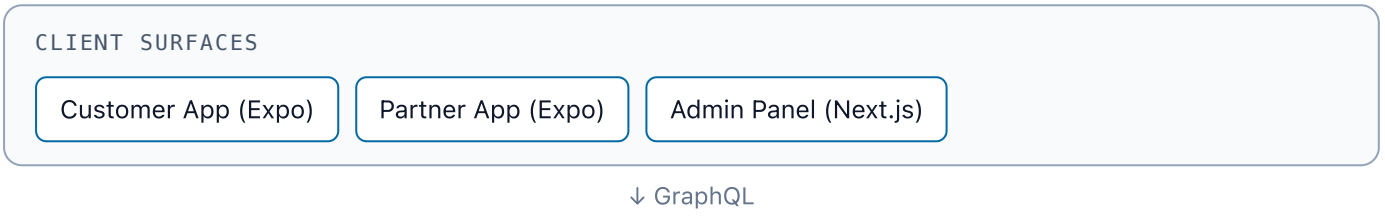
Backend Platform

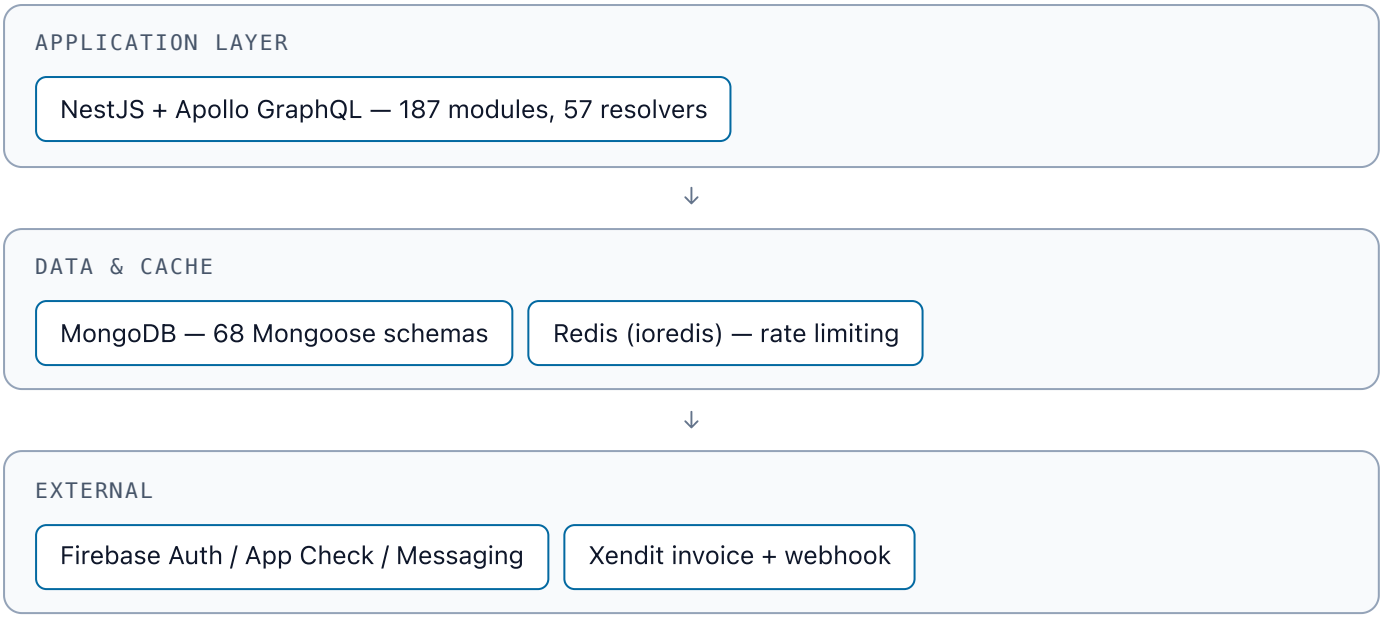
`NestJS · Apollo GraphQL`

The shared API every client talks to — one backend behind both mobile apps and the admin panel.

- 187 module directories, 57 GraphQL resolvers, 68 Mongoose schemas
- MongoDB + Redis (rate limiting/caching)
- Xendit wallet webhook, Firebase Admin SDK

4. System architecture





> The marketing website is deliberately unwired: its own repository, with no API, auth or payment connection to the platform above.

5. Technology stack

Area	Technologies
Mobile	Expo (React Native 0.83), expo-router, react-native-maps
Web	Next.js (admin + website), React 19
Backend	NestJS 11, Apollo Server / GraphQL, Mongoose 9
Data & Cache	MongoDB, Redis (ioredis)
Auth & Trust	Firebase Auth (JS SDK), Firebase App Check, Firebase Cloud Messaging
Payments	Xendit (invoice + webhook)
Verification	react-native-vision-camera, Face detection, Biometrics
Delivery	EAS Build/Update (dev/preview/production)

6. Capabilities

Customer

- Provider discovery on a map, booking, and order tracking
- In-app wallet with Xendit-backed top-ups
- Push notifications via Firebase Cloud Messaging

Partner / Provider

- Branch-scoped staff device registration and approval
- Camera-based verification (face detection, biometrics) for KYC
- Booking, POS, and online-order handling modules

Platform & Access

- Server-authoritative platform-fee, booking-availability, and booking-policy modules
- Permission/role modules enforced at the API layer, not just the UI
- Promotions and voucher redemption as first-class backend modules

Admin Operations

- 29 admin routes mirroring backend module boundaries (merchants, orders, wallets, promotions, campaigns)
- Booking-policy and platform-fee configuration UI
- Wallet and KYC review tooling

7. Engineering deep dives

Staff device approval: how access is gated per branch

1. A staff device registers itself against a specific branch
2. An owner/admin approves or blocks it — status lives on the device record, not the client
3. Each authorization check is cached for 5 minutes to avoid hammering the DB on every request
4. A device is gated to a single active session — deactivating it can't be silently bypassed by the app still holding an old token
5. A backfill routine (onModuleInit) normalizes devices created before the status/session fields existed, so old rows don't fall through the gate unprotected

Wallet top-up: how a payment callback credits exactly once

1. Client requests a wallet top-up; a pending top-up intent is recorded with its own external ID
2. Xendit hosts the actual invoice/payment collection
3. Xendit calls back to POST /webhooks/xendit with an x-callback-token header
4. The token is compared with SHA-256 + a timing-safe comparison — no early-exit string compare that could leak timing information
5. On PAID/SETTLED, the amount and currency are validated against the stored intent before anything is credited

6. The credit itself goes through one transactional path guarded by a unique ledger index, so a retried Xendit callback returns "already posted" instead of crediting twice

8. Repositories

Public, redacted showcases of each part of the system:

Part	Repository
Backend	https://github.com/pjDevph/lalaba-backend-showcase
Customer app	https://github.com/pjDevph/lalaba-customer-showcase
Partner app	https://github.com/pjDevph/lalaba-partner-showcase
Admin panel	https://github.com/pjDevph/lalaba-admin-showcase
Website	https://github.com/pjDevph/lalaba-website-showcase

9. Engineering decisions

Server-authoritative pricing

PROBLEM

Different clients could otherwise compute inconsistent totals, platform fees, and booking capacity.

DECISION

Centralize fee, voucher, availability, and booking-policy rules in dedicated backend modules (platform-fee, booking-availability, booking-policy).

IMPACT

The backend is the single source of truth for money and capacity — no client can under- or over-charge by recomputing a total locally.

Branch-scoped device approval

PROBLEM

Staff devices and permissions differ by branch, and a lost/shared device is a real access risk in a laundry-branch setting.

DECISION

Gate each device to a single active session server-side, cache the auth check for 5 minutes, and scope grants to the approved branch.

IMPACT

A device can't silently stay authorized after being deactivated, and permissions can't be inferred from UI state alone.

10. System scale

- 5 independent repos
- 187 backend modules
- 57 GraphQL resolvers

- 68 Mongoose schemas
- Multi-env EAS

> These figures describe the whole system. His own contributions are the areas of ownership listed in this section.

11. Evidence, status and limitations

- This document describes architecture and behaviour only. It contains no credentials, internal endpoints or customer data.
- No performance, revenue or user-count figures are published for this project.
- The system is a small-team project; the system counts above describe the whole backend.

Sources:

- Case study: https://pjdevph.vercel.app/projects/lalaba
- Backend: https://github.com/pjDevph/lalaba-backend-showcase
- Customer app: https://github.com/pjDevph/lalaba-customer-showcase
- Partner app: https://github.com/pjDevph/lalaba-partner-showcase
- Admin panel: https://github.com/pjDevph/lalaba-admin-showcase
- Website: https://github.com/pjDevph/lalaba-website-showcase